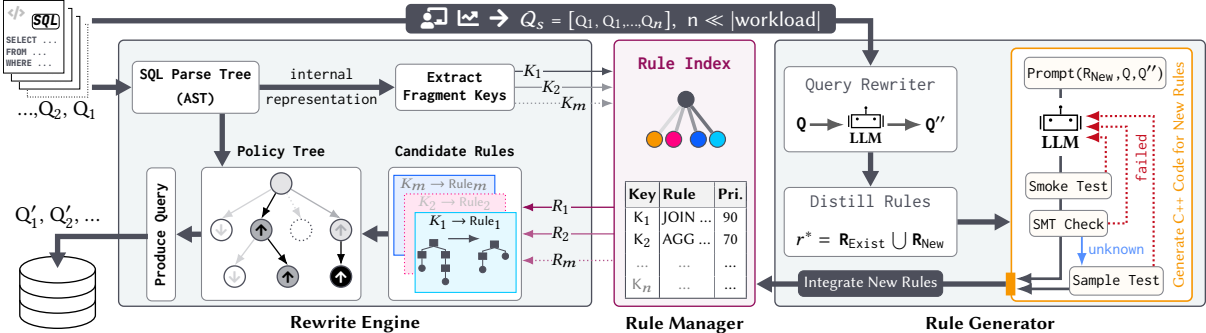

SQL
SELECT ...
FROM ...
WHERE ...
..., Q2, Q1
Qs = [Q1, Q1,...,Qn], n ≪ |workload|
SQL Parse Tree (AST)
internal representation
Extract Fragment Keys
K1
K2
Km
Rule Index
Key | Rule | Pri.
K1 | JOIN ... | 90
K2 | AGG ... | 70
... | ... | ...
Kn | ... | ...
Policy Tree
Candidate Rules
Km → Rulem
K2 → Rule2
K1 → Rule1
R1
R2
Rm
Produce Query
Q'1, Q'2, ...
Rewrite Engine
Rule Manager
Query Rewriter
Q
LLM
Q''
Distill Rules
r* = RExist ⋃ RNew
Integrate New Rules
Prompt(RNew, Q, Q'')
LLM
Smoke Test
SMT Check
failed
unknown
Sample Test
Generate C++ Code for New Rules
Rule Generator